

TABLE I. SUMMARY OF IR SENSOR DETECTION RESULTS

No.	Actual Condition	Cup–Sensor Distance (cm)	IR Sensor Status	Remarks
1	Cup Placed	0.5	LOW	Match
2	Cup Removed	0.5	HIGH	Match
3	Cup Placed	1.0	LOW	Match
4	Cup Removed	1.0	HIGH	Match
5	Cup Placed	1.5	LOW	Match
6	Cup Removed	1.5	HIGH	Match
7	Cup Placed	2.0	LOW	Match
8	Cup Removed	2.0	HIGH	Match
9	Cup Placed	2.5	LOW	Match
10	Cup Removed	2.5	HIGH	Match
11	Cup Placed	3.0	HIGH	Match
12	Cup Removed	3.0	HIGH	Match
13	Cup Placed	3.5	HIGH	Match
14	Cup Removed	3.5	HIGH	Match
15	Cup Placed	4.0	HIGH	Match
16	Cup Removed	4.0	HIGH	Match
17	Cup Placed	4.5	HIGH	Match
18	Cup Removed	4.5	HIGH	Match
19	Cup Placed	5.0	HIGH	Match
20	Cup Removed	5.0	HIGH	Match

TABLE II. DETECTION SUMMARY

Parameter	Value
Total Mismatches	0
Detection Accuracy	100%
Detection Mismatch Percentage	0%

TABLE III. REPRESENTATIVE CALIBRATION DATA FOR SYRUP AND WATER

Target Volume (mL)	Syrup (steps)	Water (steps)
10	3020	4000
30	9060	12250
60	18120	24700
90	27400	37100
120	36240	49450

TABLE IV. REPRESENTATIVE DISPENSING ACCURACY RESULTS FOR SYRUP

Target Volume (mL)	Mean Output Range (mL)	Standard Deviation Range (mL)
10	9.9773–10.0000	0–0.1056
60	59.7857–60.0000	0–0.2474
120	119.6667–120.0000	0–0.5774

TABLE V. OUTPUT VOLUME AT DIFFERENT TUBING LENGTHS FOR 4000 STEPS

Tubing Length	Observed Output Range (mL)
25 cm	9.5–10.0
30 cm	7.0–8.0
35 cm	5.0–6.5

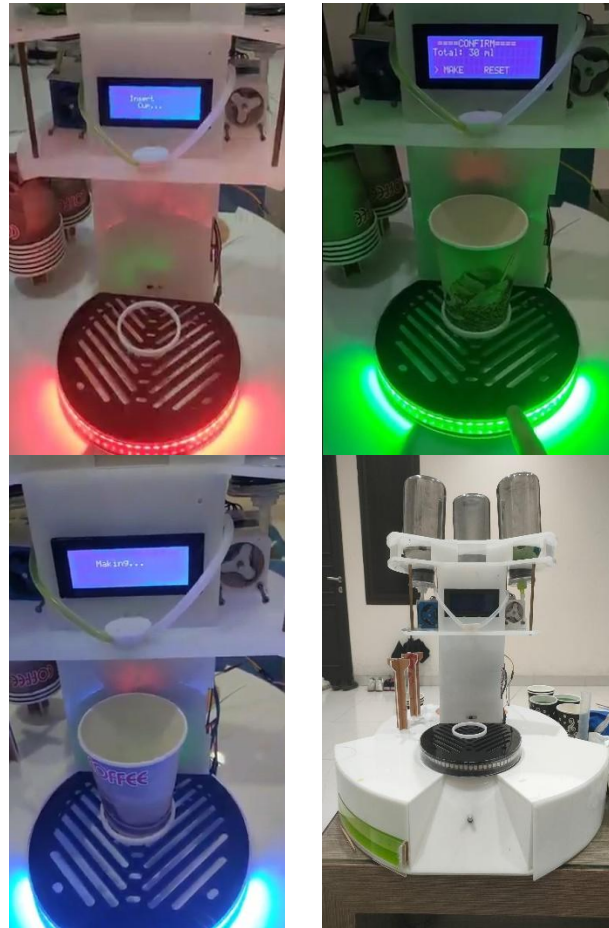


Fig. 1. Appearance of Automatic Drink Dispensing System